

Active-site-constrained structure-guided hypotheses for six uncharacterised enzymes of *Mycobacterium tuberculosis*

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Abstract

Roughly a quarter of the *Mycobacterium tuberculosis* H37Rv proteome is still annotated as “hypothetical”, and many of these conserved-hypothetical genes are predicted enzymes whose substrate and chemistry remain unassigned. Transferring function from structural matches has become routine: proteome-wide search assigns candidate folds cheaply, and several bacterial proteomes, including a mycobacterial one, have been re-annotated this way. Two things such transfers rarely test are whether the candidate can actually perform the chemistry, and whether the function proposed is not already carried elsewhere in the genome. Here we make both conditions of acceptance rather than remarks. A structure-guided pipeline (ESM-Fold modelling, Foldseek/TM-align template search, profile-profile validation with HHpred, three-dimensional active-site mapping cross-validated with AlphaFold3, transposon-derived essentiality, and purifying-selection analysis across $\sim 250,000$ sequenced *Mycobacterium tuberculosis* genomes) is applied to six conserved hypothetical proteins flagged as putative novel enzymes under two rejection criteria: the catalytic constellation must be geometrically competent, measured atom by atom on the superposed template, and the proposed function must not already be attributed to another locus (a “fold-paralogue” safeguard). Three targets emerge as robust novel enzymes with a structurally competent active site: Rv1118c, a circularly permuted NlpC/P60 amidase with an assembled Cys234–His97–Glu115 triad acting on an N-acyl-fatty-amide bond; Rv3577, a binuclear metallo- β -lactamase-fold hydrolase of the UPF0173/UlaG family with an unassigned substrate; and Rv2492, a thymidylate-synthase-superfamily (hydroxy)methyltransferase whose specificity residues point to a deoxyuridylate-type substrate rather than the cytidylate substrate suggested by its closest structural template. A fourth, Rv2558, is a cofactor-independent ABM monooxygenase with a single competent pocket but no demonstrable substrate (weaker candidate). A fifth, Rv3196, is rejected and reported as a false positive: its template match covers only the N-terminal scaffold of a YcaO heterocyclase, none of the catalytic domain, an instance of the normalised-TM-score pitfall a fold-level transfer would have propagated. The sixth, Rv3839, is a heme-homeostasis regulator independently characterised while this work was in progress, an a-posteriori confirmation of one structure-guided hypothesis. All six genes are non-essential in vitro yet their catalytic residues are essentially invariant across the sampled genomes, the signature of maintained, functional enzymes. The single loss of function we detect supports that reading: Rv1118c is disrupted by a fixed frameshift throughout *M. africanum* lineage 6 and is a pseudogene in *M. leprae*, both undergoing reductive evolution. Four of the six proteins are detected in public mass-spectrometry compendia. We provide testable biochemical predictions for each target, including the one we reject.

Keywords: *Mycobacterium tuberculosis*, dark enzymes, hypothetical proteins, structure prediction, active-site mapping, functional annotation

1 Introduction

The complete genome sequence of *Mycobacterium tuberculosis* H37Rv was published more than twenty-five years ago [1], yet roughly a quarter of its 3906 protein-coding genes are still annotated as “hypothetical” or “conserved hypothetical” [2], with no assigned biochemical function. This is more than a bookkeeping problem: among these genes are predicted enzymes whose activities could illuminate mycobacterial physiology, host adaptation, and the metabolism of a leading cause of death from a single infectious agent. Closing the gap by experiment alone is slow, so computational prioritisation that produces concrete, testable hypotheses is valuable.

Two developments make a systematic structure-guided re-annotation feasible. First, single-sequence and evolutionary-scale folding methods such as ESMFold predict atomic-resolution models for individual proteins in seconds, without a multiple-sequence alignment [3]. Second, fast structural search with Foldseek [4] and pairwise structural alignment with TM-align [5] allow a predicted model to be queried against the entire Protein Data Bank, recovering remote homologies that sequence-based tools (Pfam, BLAST) miss in the “twilight zone” of < 25% identity. Together they turn a hypothetical protein into a candidate fold, and sometimes a candidate function.

This power comes with a characteristic failure mode. A structural-similarity score can be high for the wrong reason: the TM-score is commonly normalised by the length of the shorter chain, so when a small query matches only a sub-domain of a large, multi-domain enzyme, the normalised score is inflated even though no catalytic machinery is shared. A second pitfall is the “fold paralogue”: many enzyme folds are reused across a genome, so a template match to, say, an RNase Z or a glyoxalase II does not license that label if the activity is already carried by a different, properly annotated locus. Both assign a plausible but wrong function that survives superficial review; guarding against them requires going beyond the global score, down to the active-site residues, and checking the candidate function against the rest of the genome.

Genome-scale structural annotation of the *Mycobacterium tuberculosis* proteome is not itself new: fold-recognition modelling reassigned folds to ~70% of the H37Rv genome [6], and structure-aware literature curation has more recently updated the function of hundreds of poorly characterised genes [7]. What these efforts establish at the level of the fold, however, they seldom resolve at the level of the catalytic site, and they do not systematically guard against the two failure modes above. Our own targets make the gap concrete. All six appear in the supplementary tables of Modlin et al. [7], and for four of them the template search agreed with ours: Rv3577 reached the metallo- β -lactamase CATH family and a UPF0173 metal-dependent hydrolase (3x2x), Rv2492 the thymidylate-synthase family together with a predicted UMP-binding pocket, Rv3196 the same TruD structure (4bs9) that we show below to be a fold trap, and Rv1118c the same 3kw0 and 2if6 templates we use here. Yet none of the six was reassigned: in that study’s released annotation all six keep their legacy product, four are recorded as *unassigned* and the remaining two stop at the family level. The fold was reached and not converted into a function, which is precisely the step this work is about.

Since then, structural search has become routine rather than remarkable. Foldseek has made proteome-wide structural comparison cheap and transferring functional labels from the hits a genre of its own, applied to the neighbouring pathogen *Mycobacteroides abscessus* (adding 374 and refining 885 GO annotations [8]) and to kinetoplastids, microsporidia and environmental metagenomes. Detecting a catalytic site by projecting a template’s residues onto a model is likewise mature [9], now run at AlphaFold-database scale through explicit geometry [10] or learned local-structure embeddings [11]. We therefore claim neither the structural search nor the template-based site check as our own.

What is still missing from that body of work is not a tool but a discipline. These pipelines transfer a label; almost none treats the active site as a condition the candidate must satisfy to survive, and none guards against assigning a function the genome already carries elsewhere. That failure mode has been documented for two decades in sequence space, where transfer within an enzyme superfamily is a major source of misannotation [12], and it does not disappear when the transfer is structural; recent benchmarking of machine-learning predictors on uncharacterised proteins reaches the same conclusion from another angle, that automatic methods make basic errors of reasoning which a curator avoids by consulting what is already known [13]. The contribution of this work is accordingly a set of constraints rather than a new algorithm: the site check is made blocking, the paralogue check explicit, allelic variation across a quarter of a million genomes used as an independent test of maintenance, and a candidate that fails reported as such. Concretely, each predicted model is superposed on its template, the template’s catalytic and substrate-binding residues projected onto the query, and the resulting geometry measured atom by atom, so a site is called competent only when the nucleophile, general base, metal ligands or substrate contacts fall at catalytic distances; this is complemented by two independent lines of evidence that a candidate is a real, maintained enzyme rather than a degenerating relic: transposon-mutagenesis essentiality calls [14–17] curated through MycoBrowser [2], and a purifying-selection test on the ~250,000 publicly sequenced *Mycobacterium tuberculosis* genomes (the dominant member of the *Mycobacterium tuberculosis* complex, MTBC) indexed in TB-Annotator [18], controlling for the ascertainment biases that make naive strain-weighted dN/dS misleading [19]. A mandatory fold-paralogue safeguard accompanies every assignment.

Here we apply these constraints to six conserved hypothetical proteins of H37Rv, selected during a structure-guided re-annotation as putative novel enzymes surviving an initial structure-versus-literature filter (neither already characterised nor obvious paralogue traps). Three are robust novel enzymes with competent active sites, one a weaker candidate, one a re-qualified false positive kept as an instructive case, and one was independently characterised during the study. For every target we provide a specific, falsifiable biochemical prediction.

2 Materials and Methods

2.1 Target selection

The six targets (Rv1118c, Rv2492, Rv2558, Rv3196, Rv3577, Rv3839; Table 1) were drawn from a companion genome-scale structure-guided re-annotation of the H37Rv proteome, archived at Zenodo as one functional-annotation record per coding sequence together with the pipeline that

produced them (DOI: [10.5281/zenodo.20815246](https://doi.org/10.5281/zenodo.20815246), CC BY 4.0; this concept DOI resolves to the current archive version), from which the selection funnel below is self-contained and reproducible. Of the 3906 protein-coding genes, the ~ 1053 annotated as “hypothetical” were processed through sequence-profile (Pfam, eggNOG-mapper), orthology (UniProt) and structure layers; 435 still lacked a family assignment (“dark”) after the sequence and orthology layers, 350 of these were modelled with ESMFold and searched against the PDB with Foldseek, and 46 returned a confident match to a characterized enzyme fold. Each was cross-checked manually against the literature and retained only when (i) the top structural hit was an enzyme, (ii) no Pfam domain reached the gathering threshold, and (iii) the suggested function was neither already established for the locus nor a transparent paralogue of an annotated enzyme. The cross-check resolved most of the 46 as paralogue traps, neutral fold coincidences, or loci already (or subsequently) characterized; six survived as genuinely novel enzyme hypotheses and are the subject of this study. The six candidate sequences and their best structural hits are provided as Supplementary Table S1 (with the sequences in FASTA as Supplementary Data).

2.2 Structure prediction and template search

Per-residue models were generated with ESMFold [3] through the public `foldSequence` API (<https://api.esmatlas.com>); mean pLDDT ranged from 85 to 97 and all reported catalytic residues lay in high-confidence regions (pLDDT ≥ 90). pLDDT is reported throughout on the 0–100 scale. Crystallographic templates were retrieved from the RCSB PDB. Remote structural similarity was assessed with Foldseek [4] (`easy-search`, TM-align mode) and TM-align [5]. *Two distinct structural searches are reported and should not be compared directly* (full definitions in Supplementary Note S6.3): a genome-scale screen against the whole PDB (hit, *E*-value and TM-score in Supplementary Table S1) and a dedicated per-target superposition against the single downloaded template, used for active-site mapping and quoted in Results. We further distinguish the query-normalised TM-score (`qtmscore`) from the target-normalised score (`ttmscore`): a high `qtmscore` with a low `ttmscore` flags a partial, sub-domain match (see Rv3196). Remote homology was independently corroborated by profile–profile comparison with HHpred [20, 21] (HHblits multiple-sequence alignment over UniRef30, searched against PDB_mmCIF70, Pfam-A and SCOPe70); top hits with their probabilities, *E*-values and aligned-column counts are reported in Supplementary Table S2.

Software versions and access. The Foldseek build was 427df8a; HHpred was run through the web interface of the MPI Bioinformatics Toolkit (<https://toolkit.tuebingen.mpg.de/tools/hhpred>). AlphaFold3 predictions were obtained from the AlphaFold Server in June 2026 (the apo cross-validation jobs, and the Rv3577 metal co-folding jobs, 2×Zn and 2×Fe, on 9 June 2026). All web and API queries were performed in June 2026; the analysis scripts, predicted models and tool invocations are archived at Zenodo (DOI [10.5281/zenodo.21715010](https://doi.org/10.5281/zenodo.21715010)).

2.3 Active-site mapping

For each competent candidate, the predicted model was superposed onto a ligand-bound or catalytically annotated template. Superposition used either Foldseek structural alignment or a sequence-guided rigid fit (BLOSUM62 pairwise alignment of equivalent residues, followed by

Kabsch/SVD superposition on the conserved core with iterative outlier rejection at a 4 Å cutoff). The template’s catalytic, metal-coordinating or substrate-contacting residues were then projected onto the query, and inter-atomic distances (donor–donor, Cys S γ to His N, residue side chain to bound ligand) were measured directly on the model. A site was scored competent only when these distances fell in the canonical catalytic range. Circular permutation was explicitly considered: the catalytic partner can lie far from the nucleophile in sequence while being adjacent in space. To gauge model-dependent uncertainty, each competent site was independently re-folded with AlphaFold3 (AlphaFold Server) and the same inter-residue distances re-measured; the two predictors and their per-residue confidences are compared in Supplementary Table S3, and active-site distances are read with this inter-method spread in mind. For Rv3577, the candidate metal site was tested further by co-folding the chain with two metal ions (independent Zn and Fe AlphaFold Server jobs) and measuring the metal–ligand and metal–metal distances on the holo model (Supplementary Table S4). Because a metal-templated prediction cannot by itself exclude that the ions were placed on an incidental histidine cluster, the site was also assessed by an orthogonal, template-free and metal-free predictor: BioMetAll [22] (v1.0, minimum three coordinating residues), which scores metal-binding sites from backbone preorganisation alone, was run blind on the apo ESMFold model.

2.4 Essentiality

In-vitro essentiality was taken from MycoBrowser [2], which consolidates the saturating Himar1 transposon-mutagenesis screens of Sasseti et al. [14], Griffin et al. [15], DeJesus et al. [16] and Minato et al. [17]. We report the consolidated category and note any documented in-vivo or conditional induction.

2.5 Selection on catalytic residues

For each catalytic residue the exact H37Rv codon coordinates were computed on NC_000962.3 (strand verified by translation), and all variants at these positions retrieved from TB-Annotator [18] (release indexing 250,724 *Mycobacterium tuberculosis* genomes) and classified as synonymous or non-synonymous by local re-translation (0-based SPDI coordinates, verified position by position, `ref = genome[pos]`). Because strain-weighted dN/dS is dominated by lineage-marker fixation and is therefore an ascertainment artefact [19], we rely on two unbiased readouts instead: the allele frequency of non-synonymous changes at each catalytic codon, and their presence or absence among lineage-defining markers. A count-based depletion test against matched local control codons (± 10 residues, Fisher exact) was also run for each target but proved uninformative at our sampling depth (see Discussion), so no claim rests on it. The complete per-codon variant tables for every catalytic and metal-ligand residue (SPDI position, mutation, synonymous status, strain count and lineage-marker flag) are provided as Supplementary Table S5, so the selection statements are reproducible independently of TB-Annotator; none of the 60 catalytic/ligand positions is a lineage-defining marker.

2.6 Fold-paralogue safeguard

Before any named function was retained, we verified that the activity was not already attributed to a different H37Rv locus. When a template implied a function held elsewhere in the genome (e.g. RNase Z, glyoxalase II, heme oxygenase), that label was withheld and only the fold-level family was claimed.

2.7 Figures

Predicted folds were ray-traced in PyMOL, coloured by pLDDT, with active-site residues shown as sticks coloured by catalytic role; active-site geometry panels and the synthesis plate were assembled in Matplotlib. All figures are rendered at ≥ 600 dpi in PDF and PNG.

3 Results

Table 1 summarises the six targets and their verdicts. Figure 1 shows the four enzymes whose active sites we mapped in three dimensions. We call a target a *robust novel enzyme* when three conditions hold together: its active site is geometrically competent on the model (nucleophile, base, metal ligands or substrate contacts at catalytic distances), its catalytic residues are conserved under purifying selection across the genome set, and the fold-paralogue safeguard is passed (the function is not already attributed to another locus). A target is *uncertain* when the fold is supported but the competent site or the activity is not established; the remaining outcomes are a re-qualified false positive and a regulator characterised independently of this work. We then discuss each target in turn.

Table 1: The six conserved hypothetical proteins, their closest structural template, and the consolidated verdict. All are non-essential in vitro. NE: non-essential; ABM: antibiotic-biosynthesis-monooxygenase; DRI: Domain Related to Iron.

Locus	aa	Template (PDB)	Fold / family	Verdict
Rv1118c	286	3kw0	permuted NlpC/P60 (YaeF/YiiX)	Robust novel enzyme; competent Cys-His-Glu triad; N-acyl amidase
Rv3577	288	3bv6	metallo- β -lactamase (UPF0173/UlaG)	Robust novel enzyme; binuclear metal site; substrate unassigned
Rv2492	250	5b6d	thymidylate-synthase superfamily	Robust novel enzyme; competent site; (deoxy)uridylate-type substrate
Rv2558	236	3f44	ABM monooxygenase (tandem)	Candidate (uncertain); single His124 pocket; cofactorless quinone-MO
Rv3196	299	4bs9	ThiF/MoeB/E1 adenylyltransferase	False positive for YcaO; re-routed by HHpred to a ThiF/MoeB/E1 adenylyltransferase (TM-score trap)
Rv3839	258	5che	DUF2470/DRI	Heme-homeostasis regulator; characterised independently [23]

Dark enzymes of *M. tuberculosis*: four enzymes with mapped active sites

Rv3577 - MBL-fold metallo-hydrolase

Rv2492 - (Hydroxy)methyltransferase (TS superfam.)



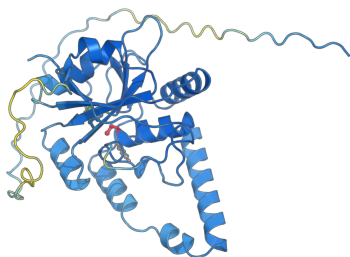
competent binuclear HxHxDH site; substrate unassigned



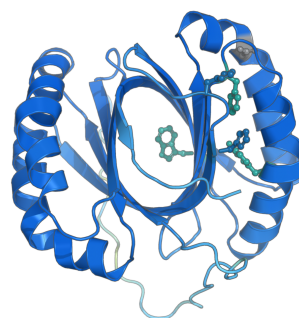
competent site (RMSD 1.5 Å); dUMP-type substrate

Rv1118c - Permuted NlpC/P60 amidase

Rv2558 - Cofactorless ABM monooxygenase



competent Cys234-His97-Glu115 triad (3.6 Å)



His124 pocket; weaker candidate

Figure 1: Synthesis of the four enzymes with mapped active sites. ESMFold models (PyMOL, cartoon coloured by pLDDT) with active-site residues as sticks. Rv3577: binuclear metallo-hydrolase site (HxHxDH). Rv2492: thymidylate-synthase-superfamily site. Rv1118c: permuted NlpC/P60 triad. Rv2558: cofactorless ABM pocket (His124). See per-target figures for geometry.

3.1 Rv1118c: a circularly permuted NlpC/P60 amidase with a competent catalytic triad

Rv1118c (286 aa, complement(1241971..1242831)) models onto 3kw0, a *Bacillus cereus* NlpC/P60 hydrolase of the circularly permuted YaeF/YiiX subfamily [24, 25]. The decisive question left open by sequence analysis was the identity of the histidine partner of the candidate nucleophile Cys234, because the three sequence-proximal histidines were all geometrically excluded. Exhaustive measurement on the model places Cys234 S γ at 3.59 Å from His97 N δ 1, a textbook catalytic-dyad distance that AlphaFold3 independently reproduces (2.93 Å; Supplementary Table S3), while the three sequence-chosen candidates (His113, His129, His160) lie 7.3–23 Å away (Figure 2). His97 sits in the N-terminal half of the chain and Cys234 in the C-terminal half, exactly the permuted topology of the YaeF/YiiX family. A carboxylate, Glu115, lies near the base on the ESMFold model (His97 N ϵ 2 to Glu115 O ϵ 1 = 3.91 Å) and may orient it as in a Cys–His–Glu triad; this third contact is, however, not reproduced by AlphaFold3 (5.9 Å; Glu115 pLDDT 64) and is reported as a candidate rather than an established triad member. A signature Tyr233 lies nearby. After TM-align superposition on 3kw0 (TM-score 0.742, RMSD 3.13 Å over 174 residues) the Cys234–His97 dyad overlays the template’s catalytic Cys–His to within 0.5–1.0 Å atom for atom.

The substrate-binding pocket (22 residues within 8 Å of Cys234 S γ) is 86% hydrophobic/aromatic, compatible with a fatty-acyl chain and incompatible with a charged peptidoglycan muropeptide. This matches the biochemistry of the permuted NlpC/P60 subfamily, which cleaves the amide bond between a fatty acid and an amino acid rather than acting as a classical peptidase [24]. Rv1118c is distinct from the five named canonical NlpC/P60 enzymes of H37Rv (Rv0024, RipA, RipB, RipD, Rv2190c), which are peptidoglycan hydrolases. Its absence from the current inventory of the family is itself informative: a 2025 review of the mycobacterial NlpC_P60 peptidases covers those five at length and does not mention Rv1118c [26], as expected for a member whose circular permutation removes the sequence signal such surveys rely on. It is non-essential *in vitro* by three concordant screens [14–16]. Its catalytic codons are essentially invariant across 250,724 genomes (Supplementary Table S5): the catalytic dyad is tightly conserved, with 7 strains (0.0028%) carrying a Cys234-disrupting change and only 3 (0.0012%) an His97 change; the candidate orienting acid Glu115 is more relaxed, its commonest non-synonymous change being E115Q (n=27, 0.011%) ahead of the carboxylate-conserving E115D (n=5), as expected for a third contact that AlphaFold3 does not reproduce; neighbouring codons tolerate substitution freely, so this targeted protection is the signature of purifying selection on a functional site rather than block conservation. Conservation extends well beyond the timescale this intra-MTBC signal can probe: Rv1118c is detected in all 53 sequenced non-tuberculous mycobacterial genomes surveyed, and remains identifiable across the wider order Corynebacteriales, in *Gordonia*, *Mycobacteroides*, *Nocardia* and *Rhodococcus* (mean 58% identity), indicating order-level constraint on a gene that intra-MTBC data alone, sampled from a single clonal population, cannot date. The gene is intact in *M. bovis*, and it is lost twice within the complex, in both cases in a genome-reduced context: it is a pseudogene in *M. leprae*, and it is disrupted throughout *M. africanum* lineage 6 (L6) by a fixed four-nucleotide deletion (H37Rv 1,242,213–1,242,216) that shifts the reading frame at codon 206 of 286, upstream of the Cys234 nucleophile. All 1,387 L6 genomes

of the companion MTBC collection carry this deletion and no genome of any other lineage does; a secondary premature stop (W263*) has since accumulated in a fifth of them, as expected once a coding sequence is no longer maintained. This loss does not appear in the frequencies above because the TB-Annotator denominator is *Mycobacterium tuberculosis sensu stricto*, which excludes *M. africanum*. That the only two losses fall in lineages undergoing reductive evolution [27] is what one expects of a dispensable but functional enzyme, and the opposite of the pattern a spurious reading frame would show.

Profile–profile comparison corroborates the assignment and sharpens the substrate. HH-pred matches Rv1118c to the permuted YaeF/YiiX Peptidase_C92 family (Pfam PF05708; PDB 3kw0, 2if6) at probability $\geq 99.7\%$, spanning the entire Cys234–His97–Glu115 triad, while canonical peptidoglycan NlpC/P60 enzymes appear only as distant matches ($\sim 88\%$). A 99.3% hit to a lipoprotein N-acylation family (LnsA-like, Pfam PF28265) and to several lipid-acting permuted members (LRAT, H-REV107) independently reinforces an N-acyl/lipid substrate, plausibly lipoprotein N-acylation.

Prediction. Recombinant Rv1118c should hydrolyse N-acyl-amino-acid or lipoprotein-mimetic amide substrates; the C234A and H97A mutants should be inactive.

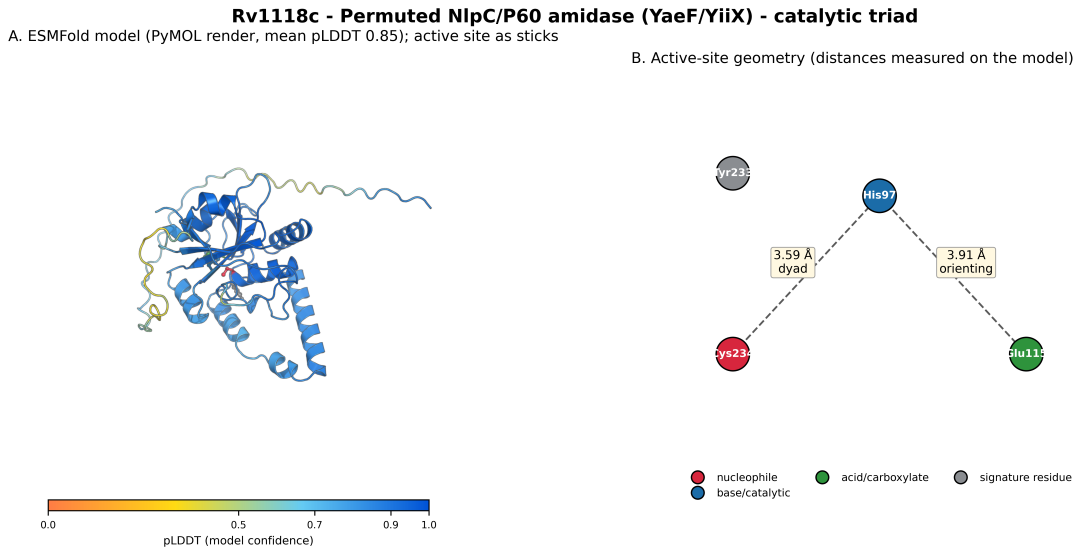


Figure 2: Rv1118c. Left: ESMFold model (pLDDT colouring) with the Cys234–His97–Glu115 triad as sticks. Right: active-site geometry; the catalytic dyad Cys234 S γ –His97 N δ 1 measures 3.59 Å, and the three sequence-proximal histidines (His113/129/160) are refuted at 7.3–23 Å. His97, in the N-terminal half, reflects the permuted topology.

3.2 Rv3577: a binuclear metallo- β -lactamase-fold hydrolase

Rv3577 (288 aa, 4019262..4020128) carries an intact metallo- β -lactamase (MBL) fold with the HxHxDH motif (His74–His76–Asp78–His79). Its template, 3bv6, is a *Vibrio cholerae* di-iron metal-dependent hydrolase of unknown substrate. Dedicated structural superposition on that template (Foldseek probability 1.00, $E = 3.4 \times 10^{-14}$, alignment TM-score 0.61; the genome-scale screen reported $E = 1 \times 10^{-12}$ for the same pair, Supplementary Table S1) projects five of the seven template metal ligands exactly onto conserved His/Asp residues of Rv3577 (His74, His76,

Asp78, His79, His235). To test whether these residues form a genuine two-metal centre rather than an incidental histidine cluster, the chain was co-folded with two metal ions on AlphaFold Server, in independent Zn and Fe jobs (top model iPTM 0.98, pTM 0.93–0.94; Supplementary Table S4). In both jobs the two ions converge on a single bridged binuclear site, 3.2–3.4 Å apart, each ion engaging three protein ligands at bonding distance (~ 2.1 Å): a three-histidine metal (His74/His76/His137) and an aspartate-plus-two-histidine metal (Asp78/His79/His235), the canonical two-metal architecture of the MBL superfamily (e.g. glyoxalase II, RNase Z; Figure 3). The geometry is reproduced essentially unchanged with Zn or Fe, consistent with the metal promiscuity of this fold, and all six ligands sit at high local confidence (pLDDT 88–98). The co-folding identifies His137 as the sixth ligand and shows that Asp141 and Cys162, although near the cleft in the apo model, do not coordinate metal.

Because metal-templated co-folding could in principle place ions on any plausible histidine cluster, we sought orthogonal, template-free support (BioMetAll, blind on the apo model) and a six-alanine specificity control (re-folded in holo): both independently return the same binuclear site as the dominant metal-compatible region of the backbone, and abolishing the six ligands abolishes metal binding at the site and relocates it to the same secondary cluster both methods had already ranked as runner-up (full geometry and probe-cluster detail in Supplementary Note S6.1–S6.2). Backbone geometry, holo co-folding and the mutant control therefore agree that the di-metal centre is genuine and specific, not an incidental histidine cluster.

Rv3577 is non-essential in vitro across four screens [14–17] but is induced in immunocompetent (BALB/c) more than immunodeficient (SCID) mice at day 21, suggesting a conditional in-vivo role. No MTBC lineage has fixed a substitution at any of the six metal ligands across 250,724 genomes (none of the six ligand codons carries a lineage marker); the most frequent non-synonymous ligand substitution (H235R) reaches only 0.0032% (8 strains), and the entire non-synonymous load across the six ligand codons sums to 47 strains (0.019%). The metal site is thus effectively invariant, an enzyme that maintains a costly binuclear centre and is therefore used. We deliberately do *not* claim RNase Z (held by Rv2407), glyoxalase II (Rv0634c/Rv2581c) or β -lactamase activity (the template is itself a non-lactamase): the defensible assignment is an MBL-fold binuclear metallo-hydrolase of the UPF0173/UlaG family (InterPro IPR050114), substrate unassigned. HHpred confirms this family-level call: the top hits ($\geq 99.8\%$) are UPF0173/UlaG metal-dependent hydrolases (PDB 2wyl, the eponymous UlaG; COG2220; several UPF0173 hydrolases) together with the template 3bv6, whereas RNase Z appears only as a lower-probability, more distant relative of the same fold, consistent with deliberately withholding that label.

The genomic neighbourhood looked like a way to narrow the substrate down, and it is not. Rv3577’s operon-like arrangement with the arsenite permease *arsB2* recalls the *E. coli ula* operon whose MBL member is the eponymous UlaG, but the same synteny is matched by 34% of randomly drawn adjacent H37Rv gene pairs (empirical $p = 0.36$) and dissolves entirely outside the genus (Supplementary Note S6.4). The adjacency is therefore genus-level co-inheritance, not a conserved functional cassette, and it carries no substrate information. We report it because the negative closes a line of reasoning that the annotation itself invites.

Prediction. Recombinant Rv3577 should bind two metal ions (ICP-MS/anomalous diffraction to identify Zn/Fe/Mn) and hydrolyse a phosphoester, oligonucleotide or lactone substrate;

alanine substitution of the ligands of either metal (His74/His76/His137 or Asp78/His79/His235) should abolish metal-dependent catalysis.

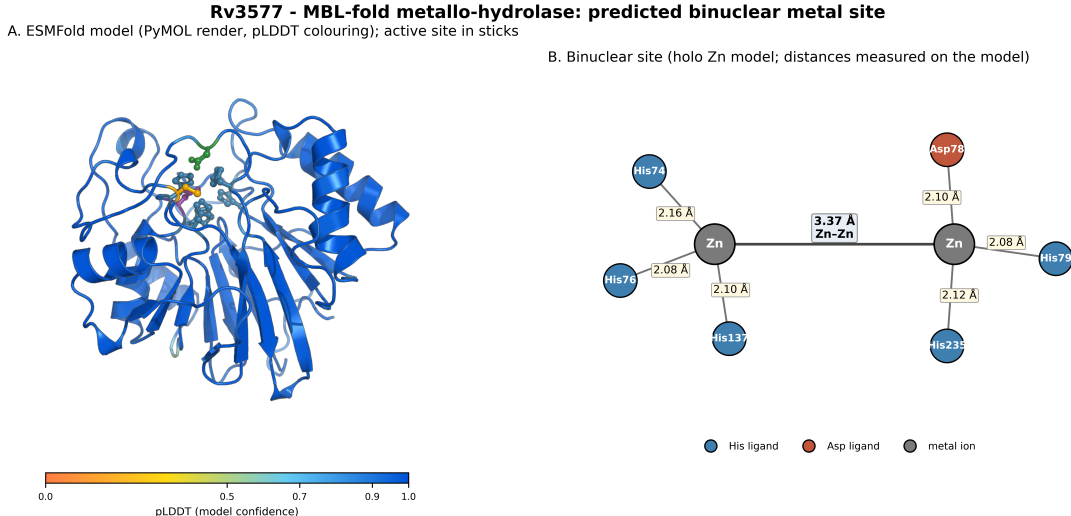


Figure 3: Rv3577. ESMFold model (pLDDT colouring) and the binuclear metal site. Co-folding with two metal ions (AlphaFold Server, Zn and Fe) places the ions 3.2–3.4 Å apart on a three-histidine metal (His74/His76/His137) and an Asp+2-His metal (Asp78/His79/His235), the canonical two-metal MBL architecture (Supplementary Table S4). Five of seven template (3bv6) metal ligands map onto conserved residues.

3.3 Rv2492: a thymidylate-synthase-superfamily enzyme with a deoxyuridylate-type signature

Rv2492 (250 aa, 2807278..2808030) adopts the thymidylate-synthase (TS) superfamily fold of MilA, a CMP (hydroxymethyl)transferase [28]. Superposition of the model on MilA bound to CMP (PDB 5b6d; 25% identity, RMSD 2.19 Å over 92 core residues) places a complete catalytic constellation at hydrogen-bonding distance of the nucleotide ligand: the nucleophile Cys151, the phosphate-binding Arg175 (2.1 Å salt bridge), the base-reading Asp178 (2.1 Å), and the ribose-binding His222/Tyr224 (1.8 Å); the site is competent, not degenerate (Figure 4). This pocket is not reported here for the first time: Modlin et al. [7] already list Rv2492 with a thymidylate-synthase CATH family and a predicted UMP-binding site (against 4g9u) whose contact residues include every position above. What that prediction did not do is assign a role to each contact, which is what turns a ligand-binding pocket into a testable chemistry, and the locus accordingly remained annotated “hypothetical protein”. The two contributions of this section are therefore the mechanistic reading of the constellation and the substrate reassignment that follows from it.

The specificity residues, however, reassign the substrate away from the CMP suggested by the closest template. MilA reads cytosine through Asp186 (MilA numbering); the experimentally established rule is that uracil-using members of the superfamily (TS, dUMP hydroxymethylase) carry an asparagine at this position [28]. Rv2492 carries Asn186 (Rv2492 numbering), the cytosine co-determinant Tyr104 is absent (gap), and Tyr72 is replaced by Phe78 (removing its hydroxyl), together pointing to a uracil base. At the sugar-discriminating position (MilA Ala176, which makes room for the ribose 2'-OH; Ser in the deoxy-preferring TS/T4-CH enzymes) Rv2492 carries

Ser176, the deoxy-type residue: the combined signature is a (deoxy)uridylylate-monophosphate, i.e. dUMP-type rather than CMP, with one-carbon chemistry on the pyrimidine C5. Profile–profile comparison places Rv2492 unambiguously in the thymidylate-synthase/dCMP-hydroxymethylase superfamily (HHpred probability 100% to the MilA template, the Pfam Thymidylat_synt family, and numerous thymidylate synthases and the T4 dCMP hydroxymethylase); the equally strong match to the essential *Mycobacterium tuberculosis* thymidylate synthase ThyA reflects superfamily kinship, not identity, Rv2492 being a distinct, non-essential locus sharing only 23% of its residues with ThyA (20% with ThyX) in global pairwise alignment.

Rv2492 is non-essential in vitro [16, 17] and is not redundant with the essential thymidylate synthases thyA/thyX (distant, distinct loci), so the TS-like signature indicates substrate and chemistry type, not functional redundancy. Its catalytic codons are strongly but not absolutely conserved across 250,724 genomes: no non-synonymous active-site variant reaches 0.1% and none is a lineage marker (the most frequent are H222N, $n = 101$ [0.040%], Y224C, $n = 33$, R175Q, $n = 30$, and D178G, $n = 23$; the nucleophile Cys151 carries only two non-synonymous strains), consistent with relaxed purifying selection on a dispensable but maintained enzyme.

We explicitly reject an earlier anti-phage-defence hypothesis, though not on the grounds we first advanced. Rv2492 is restricted to the *M. canettii*–MTBC clade (absent from all 53 non-tuberculous mycobacterial genomes surveyed in the companion re-annotation; reciprocal-best-hit orthologues only in *M. bovis* and *M. orygis*), so it is not an ancient core gene of the genus. The published record agrees: Rv2491–Rv2492 was horizontally acquired during the step-wise genesis of *Mycobacterium tuberculosis*, with reduced GC content [29], which we confirm for Rv2492 alone (51.0% versus 65.6% genome-wide, lowest 0.1% of same-length H37Rv windows); the surrounding Rv2491–Rv2494 island carries the toxin–antitoxin pair vapBC38 [30]. What argues against a defence function is the manner of conservation, not the age of the gene: the reading frame is intact and the catalytic constellation is maintained across the human and animal MTBC and *M. canettii*, whereas defence systems characteristically turn over rapidly with a patchy presence/absence pattern. We therefore retain the thymidylate-synthase-superfamily assignment, while acknowledging that comparative genomics alone cannot exclude a defence-associated role.

Prediction. In-vitro kinetics of recombinant Rv2492 (with CH₂-THF/FAD as appropriate) on dUMP, dCMP and CMP should favour the deoxy uridylylate substrate; targeted metabolomics of a Δ Rv2492 mutant should reveal a missing modified pyrimidine nucleotide.

3.4 Rv2558: a cofactor-independent ABM monooxygenase (weaker candidate)

Rv2558 (236 aa) is a tandem ABM (antibiotic-biosynthesis-monooxygenase)-domain protein with a paralogue, Rv2557 (69% identity). It models full-length onto the tandem ABM template 3f44 (TM-score 0.70); each single-domain template matches only one of its two $\sim$ 90-residue ABM lobes, which are 28–30 Å apart. Only the C-terminal lobe forms a competent pocket: His124 is the most buried histidine, lined by the aromatics Phe168 (6.8 Å) and Tyr154 (10 Å), the canonical signature of a cofactorless ABM site in which an aromatic substrate substitutes for flavin (Figure 5). The HRDH motif is not a metal site (His124–His127 = 8.9–9.5 Å across the two predictors, too open to chelate a metal).

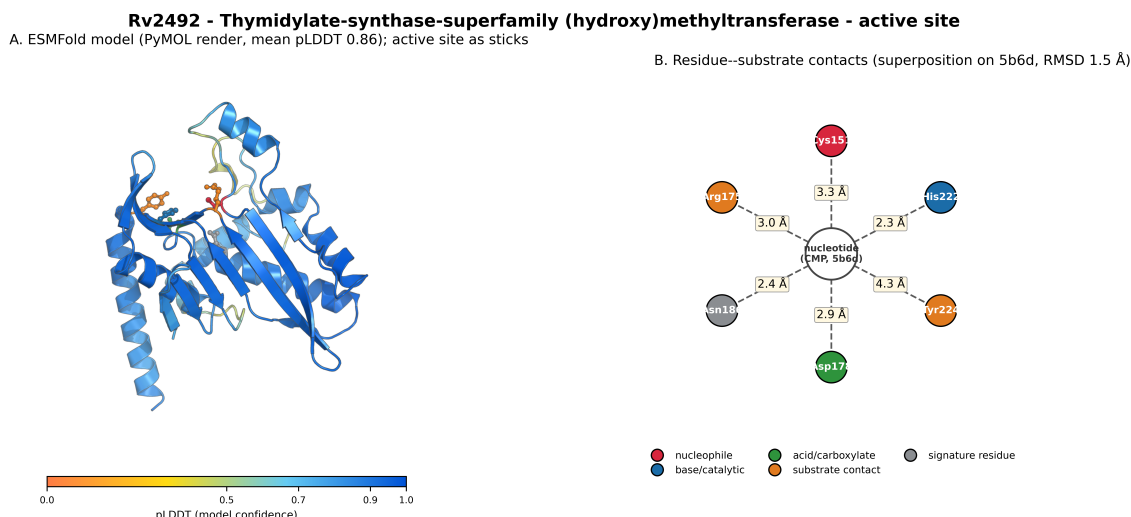


Figure 4: Rv2492. ESMFold model and contacts to the nucleotide ligand after superposition on MilA+CMF (5b6d). The catalytic core (Cys151, Arg175, Asp178, His222, Tyr224) reaches every part of the monophosphate; the base-specificity residue Asp186→Asn186 and the sugar residue Ser176 reassign the substrate to a deoxyuridylate type.

We rule out a heme-oxygenase assignment on two grounds (residue-level mapping in Supplementary Note S6.5): structurally, the IsdG/MhuD catalytic determinants are absent at their expected positions and the geometry is not heme-compatible; functionally, the fold-paralogue safeguard applies, since heme degradation is already carried by MhuD (Rv3592) [31]. The defensible assignment is therefore a cofactor-independent ABM monooxygenase of the quinone-monooxygenase family (ActVA-Orf6/SnoaB/YgiN) [32], substrate unknown. HHpred supports this sub-type ($\geq 99.9\%$ to the ActVA-Orf6/YgiN/anthrone-monooxygenase family, with the IsdG/MhuD subfamily absent from the high-probability hits), its top hit pointing to an aromatic/polyketide substrate. Both paralogues are non-essential in vitro across four screens; a targeted knockout of the carbon-starvation-inducible Rv2557–Rv2558 pair is moreover dispensable for long-term survival under carbon starvation and for virulence in SCID mice [33]. Both are under purifying selection (per-distinct-variant $pN/pS = 0.55$ for Rv2558, 0.63 for Rv2557, computed over the full MTBC set indexed in TB-Annotator rather than the *Mycobacterium tuberculosis* sensu stricto subset used elsewhere, and reported descriptively rather than as significance tests, per the count-based caveat in the Discussion); the C-terminal site residues (His124, His127, Trp139, Tyr154) are near-invariant across the 250,724 genomes, contrasting with a more variable N-terminal lobe. We label this target uncertain not on its fold or sub-type, both well supported by the structure and profile-profile evidence, but on its substrate and activity, which remain undemonstrated [34].

This weakest assignment sits, plainly stated, on the strongest protein of the six. Rv2558 is the most constrained: a single segregating site over the companion collection (145,209 genomes at the date the conservation layer was computed), with no missense change at all, where the other five carry five to fourteen such sites. It is also the most abundant, detected in 14 of 16 proteomics datasets at the 80th abundance percentile, and present in all 53 non-tuberculous mycobacterial genomes surveyed. Its eggNOG orthologous group (COG1359) carries a starvation-response

annotation, consistent with the carbon-starvation induction reported for the Rv2557–Rv2558 pair [33]. None of this makes the monooxygenase hypothesis more likely, but it does establish that the uncertainty is ours, not the cell’s: this protein is manufactured in quantity and protected from change, and its chemistry, not its reality, remains open.

Prediction. Recombinant Rv2558 should oxidise model quinone/aromatic substrates in an O₂-dependent, cofactor-free reaction, with His124Ala abolishing activity and no Soret-band heme binding.

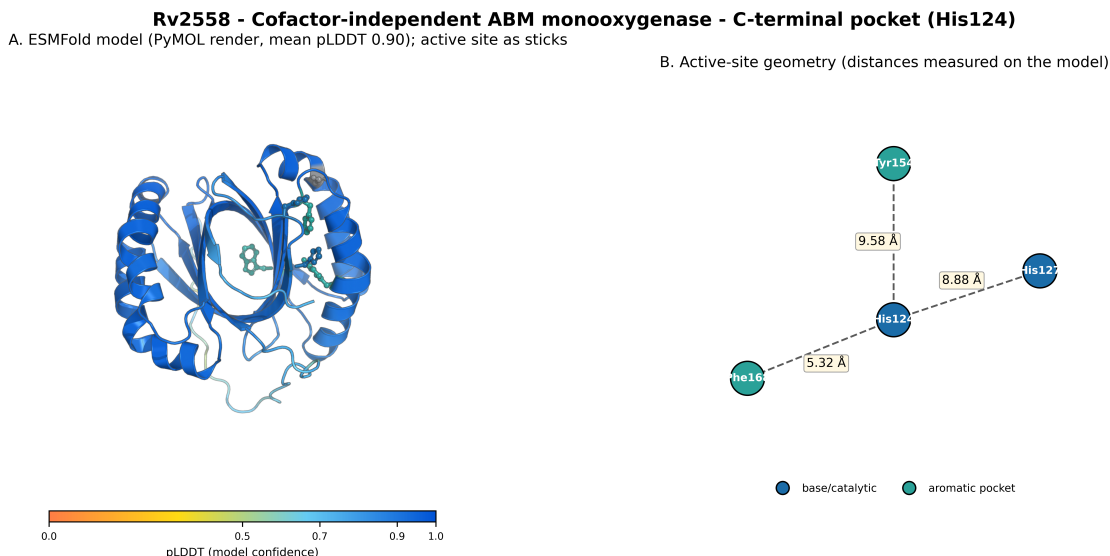


Figure 5: Rv2558. ESMFold model with the C-terminal ABM pocket centred on His124 and bordered by Phe168 and Tyr154. The HRDH histidines are too far apart to coordinate a metal, arguing against a heme-oxygenase assignment; the heme-degrading function is already held by MhuD.

3.5 Rv3196: a false positive and a cautionary case

Rv3196 (299 aa) was flagged by a strong hit to 4bs9, the cyanobactin heterocyclase TruD (a YcaO-family cyclodehydratase) [35], at a TM-score of 0.63 in the genome-scale screen (Supplementary Table S1) and a query-normalised TM-score of 0.72 on dedicated superposition. Active-site mapping reclassifies it as a false positive (Figure 6). The structural alignment covers only TruD residues 3–321, i.e. the N-terminal scaffold domains, and stops two residues before the YcaO catalytic domain (323–781): 319 of the 781 template residues, or 41% of the target, against a target-normalised TM-score of 0.33. None of the YcaO catalytic residues (the Glu/Arg pair and Mg²⁺/ATP-binding residues of the “new ATP-binding motif” [36]) lies in the aligned region, and the Rv3196 CxxC pair (C208/C211) does not correspond to the YcaO zinc site but matches the CxxC zinc-binding motif of ThiF/MoeB adenylyltransferases (see below). A RiPP role is independently improbable: H37Rv contains no precursor peptide cassette, no maturation protease, and no TfuA or second YcaO gene; the genomic neighbourhood is an ordinary gene-dense region.

Rv3196 is nonetheless a genuine, conserved enzyme: it is non-essential in vitro across four screens, conserved in > 99.98% of 250,724 genomes and in *M. canettii*, and its candidate site

is under purifying selection. Profile–profile comparison re-routes it to a positive family rather than leaving it indeterminate: its top HHpred hits ($\geq 99.9\%$) are standalone adenylyltransferases (MccB, PaaA) and the broader ThiF/MoeB/E1-like adenylation superfamily, within which the CxxC pair corresponds to the MoeB-type zinc-binding motif; cyanobactin YcaO enzymes appear only over the shared adenylation scaffold, never over their catalytic domain. Rv3196 is therefore not a heterocyclase but a probable ThiF/MoeB/E1-family adenylyltransferase of undetermined substrate, a worked example of the normalised-TM-score trap the active-site step is designed to catch.

Rv3196: a structural false positive (normalised-TM-score trap)

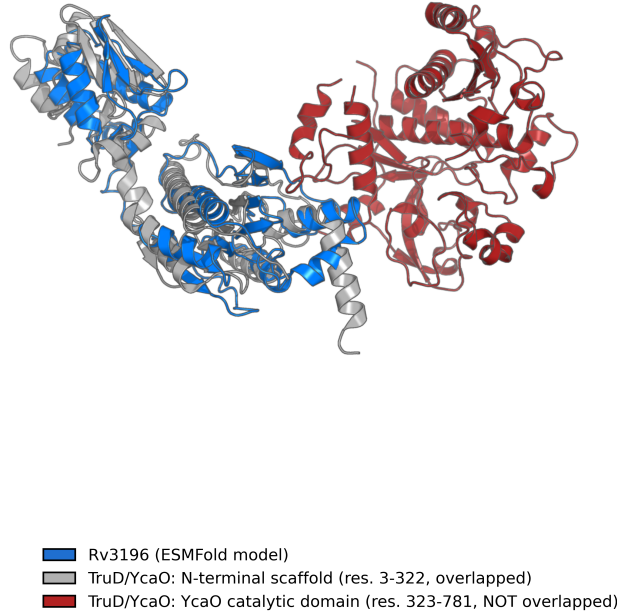


Figure 6: Rv3196 as a structural false positive. The model (blue) superposes on the N-terminal scaffold of the TruD heterocyclase (grey) but not on the YcaO catalytic domain (red), which carries the ATP/Mg cyclodehydratase machinery. The high query-normalised TM-score (0.72) is an artefact of normalising by the small protein: the alignment covers only 41% of the target, and the target-normalised score is 0.33. Rv3196 is re-qualified as a nucleotide-binding-fold enzyme of indeterminate function.

3.6 Rv3839: independent validation of the approach

Rv3839 (258 aa) carries a DUF2470/DRI (“Domain Related to Iron”) domain, hit through the GluTR–GBP complex template 5che [37]. The fold-paralogue safeguard is passed: Rv3839 is not a copy of the genuine glutamyl-tRNA reductase (hemA, Rv0509) but shares the regulatory DUF2470 domain of the GluTR-binding protein, a heme-dependent regulatory module also found in HugZ [38, 39]. It lies immediately upstream of bacterioferritin bfrB (Rv3841) in an operon

with the transcriptional regulator Rv3840, the expected context for heme/iron homeostasis. While this study was in progress, Quirk et al. [23] established that the Rv3839–Rv3840 operon down-regulates endogenous heme biosynthesis under nitric oxide and iron-limitation stress, with Rv3839 as the principal driver: a “RefSeq-lagging” case, an uncharacterised annotation whose function the structure-guided hypothesis correctly anticipated (Discussion).

3.7 Do these proteins exist? Mass-spectrometric evidence

A reasonable objection to any structure-guided assignment is that the protein may never be made. We therefore checked all six against PaxDb 5.0, which aggregates 16 independent *Mycobacterium tuberculosis* proteomics datasets [40]. Four are detected: Rv2558 in 14 of the 16 datasets and abundantly (264 ppm, 80th abundance percentile of the proteome), Rv1118c in 11 (44 ppm, 48th), Rv3577 in 7 but faintly (0.55 ppm, 5th), and Rv3839 in a single dataset (0.18 ppm, 2nd). Rv2492 and Rv3196 are detected in none.

That last observation should be weighed, not waved away: Rv3839, the one target with an experimentally established function [23], sits in the bottom 3% of the detectable proteome, in one dataset out of sixteen. Non-detection in shotgun proteomics therefore tracks abundance and physico-chemical amenability, not functional reality: low-abundance, conditionally expressed or peptide-poor proteins are systematically under-sampled. Mass spectrometry thus confirms expression for four targets, is uninformative for two, and cannot by itself argue against any of them, though it does make Rv2492 and Rv3196 the two worth an expression-level experiment first.

4 Discussion

Applied to six conserved hypothetical proteins, these constraints produced a clean partition: three robust novel enzymes with competent, conserved active sites (Rv1118c, Rv3577, Rv2492), one weaker candidate (Rv2558), one re-qualified false positive (Rv3196), and one independently confirmed regulator (Rv3839). The value of the approach is not the template search, which is now routine, but the discipline imposed after it. Four checks did the discriminating work, and a fifth earned its place by returning nothing.

First, three-dimensional active-site mapping repeatedly overturned or refined the sequence-level expectation (Results: the permuted His97 of Rv1118c, the substrate reassignment of Rv2492, the demotion of Rv3196) in ways a global similarity score could not have called.

Second, the fold-paralogue safeguard prevented two plausible over-claims, a second heme oxygenase for Rv2558 and an RNase Z/glyoxalase II label for Rv3577 (Results), both held elsewhere in the genome; withholding a named function in favour of a fold-level family is less satisfying than a headline assignment, but it is what the evidence supports.

Third, the population-genomic signal converges with the structural one, but only when read as allele frequency rather than variant counts. At a sampling depth of $\sim 250,000$ genomes, rare deleterious non-synonymous singletons occur at nearly every codon, so a depletion test at catalytic versus matched local control codons is uninformative (pooled Fisher odds ratio 1.7, not significant) and strain-weighted ratios are dominated by lineage-marker ascertainment

[19]. Purifying selection is instead evident in two unbiased ways: no non-synonymous change at any catalytic or metal-ligand residue is fixed within a lineage, and active-site residues carry non-synonymous variants in at most $\sim 0.04\%$ of genomes (the two cysteine nucleophiles in only 0.0008% and 0.0028% ; below 0.01% at 17 of the 22 catalytic codons examined), against 0.08% and 0.20% for *synonymous* changes at those same codons and their immediate neighbours. The signal is thus the non-synonymous-versus-synonymous contrast within a fixed neighbourhood, not raw non-synonymous burden, which is dominated by a single Rv2492 variant (H222N, $n = 101$) and does not separate catalytic from control codons. For genes non-essential in vitro, this frequency-level conservation argues for selected, if conditional, roles, consistent with the in-vivo induction of Rv3577 and the stress-responsive regulation of Rv3839.

The one loss of function we observe (Results: Rv1118c, complete in both cases rather than partial) points the same way. A gene dispensable in vitro, invariant across a quarter of a million genomes, and then abandoned outright in the two genome-reduced branches of the group behaves like a function held under conditional selection and released when the niche no longer requires it; a spuriously annotated reading frame would instead degrade gradually and everywhere, not cleanly and twice.

Fourth, mass spectrometry (Results) served as an existence check, not evidence of function: Rv3839, the one target with an independently established function, is itself detected in only one of sixteen datasets. Non-detection measures abundance and peptide amenability, not functional reality, and should order the experimental queue rather than argue against an assignment.

The fifth check is the one that returned nothing, reported for that reason: the syntenic arrangement around Rv3577 (Results) failed its own null model. A method advertising rejection criteria must publish the cases where its own controls remove an attractive inference, or the criteria are decorative.

The principal limitation is that every assignment rests on predicted models and structural homology; no catalytic activity has been demonstrated biochemically. A second, independent predictor (AlphaFold3) cross-validates every confident geometry to within $0.1\text{--}0.65\text{ \AA}$ and flags the single method-dependent one, Rv1118c Glu115 (Supplementary Table S3); for Rv3577 a further, metal-free and template-free predictor (BioMetAll) and a six-alanine knockout independently corroborate the same site (Results; Supplementary Table S4). Two further conservation tests (family-wide MSA depth; masked protein-language-model scoring) corroborate Rv1118c, Rv2492 and Rv3577, both correctly failing Rv2558, and an M-CSA cross-validation recovers Rv2492's catalytic core at 26% identity (Supplementary Note S7). Substrate identity remains open for Rv3577 and only narrowed for Rv2492 and Rv2558; we have stated a falsifiable prediction with a catalytic-mutant control for each target, since substrate-level certainty requires experiment. Rv3839 is an encouraging a-posteriori proof of concept, but a single case does not measure the pipeline's reliability, and the six targets are not a random sample but the survivors of a manual filter (46 confident enzyme-fold hits triaged to six); their breakdown illustrates the discipline's behaviour on hand-selected cases, not its precision or recall, which would require a held-out benchmark.

5 Conclusion

Structural search is no longer the limiting step in functional annotation; what it lacks is a set of conditions a candidate must satisfy before a label is transferred. Imposing four of them, a blocking active-site check, an explicit fold-paralogue safeguard, allelic variation across a quarter of a million genomes as a test of maintenance, and the obligation to report the candidates that fail, converts “hypothetical” annotations into concrete, testable enzyme hypotheses, removing the structural false positives a similarity score alone would retain. For *Mycobacterium tuberculosis* H37Rv it identifies three robust novel enzymes (a permuted NlpC/P60 N-acyl amidase Rv1118c, a binuclear MBL-fold metallo-hydrolase Rv3577, and a TS-superfamily deoxyuridylate-type (hydroxy)methyltransferase Rv2492), one weaker candidate (Rv2558), one instructive false positive (Rv3196), and one independently confirmed regulator (Rv3839). Each robust target comes with a biochemical prediction ready for experimental test. The same constraints apply directly to the remaining uncharacterised enzymes of the MTBC proteome, and cost nothing beyond the willingness to let a candidate fail them.

Data and code availability

The candidate sequences and their structural hits, the ESMFold and AlphaFold3 models (including the Rv3577 holo and site-knockout models), the analysis scripts and the per-target dossiers are deposited at Zenodo (DOI: [10.5281/zenodo.21715010](https://doi.org/10.5281/zenodo.21715010), CC BY 4.0). The companion re-annotation archive from which the six targets were drawn, comprising one functional record per H37Rv coding sequence and the pipeline that produced them, is deposited separately (DOI: [10.5281/zenodo.20815246](https://doi.org/10.5281/zenodo.20815246), CC BY 4.0). Supplementary Tables S1–S5, Supplementary Note S6 (extended methodological detail) and Supplementary Note S7 (independent robustness checks) accompany this manuscript.

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Competing interests

The author declares no competing interests.

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